

ECON 351 · Focused Practice Set

Set 1 of 10

Chapter 4

Consumer Problems

2 questions in this practice set

Questions: Q1, Q2

Screenshots from original HW, sample questions, and mock midterm.

Question 1 of 2 (Practice Q1)

Consumer Problem #1 · HW1 Ex 3

Exercise 3 - Optimal Consumption

3.1 Manuel has the following utility $U(q_1, q_2) = 3q_1^{0.6}q_2^{0.4}$. What values of q_1 and q_2 maximize his utility?

3.2 Now suppose prices are $p_1 = 5$ and $p_2 = 10$ and that Manuel has \$100. Write Manuel's budget constraint and graph it with q_1 in the x-axis. What is the slope? And the intercept? What is the maximum amount of q_1 he can purchase? And the maximum amount of q_2 he can purchase?

3.3 Next, we will find out the best use Manuel could make of his money, given his preferences. Based on Manuel's preferences, let's figure out the ratio between q_1 and q_2 . What is the ratio of the two marginal utilities (MRS)? (remember that MU_1 is in the numerator)

3.4 Now we have the ratio of how Manuel trades both goods. We will equalize it to the ratio of both prices. That is, how Manuel trades both goods needs to equal how the market trades them. In the optimal bundle, how do the marginal utilities relate to market prices?

3.5 Finally, use the previous relationship to substitute in the budget constraint and find the optimal quantities of q_1 and q_2 , which we denote by q_1^* and q_2^* .

Question 2 of 2 (Practice Q2)

Consumer Problem #2 · Sample Q26

Question 26:

There are two goods, apples and bananas. The price of apples is $P_A = \$2$, and the price of bananas is $P_B = \$3$. A consumer has \$120 to spend, and his utility function is $U(A, B) = 2A^2B^3$.

a) Write the expression that represents the consumer's budget constraint. Graph the budget constraint, and label it "BC" (apples should be on the horizontal axis). What is the slope of the budget constraint?

b) Find the consumption basket that maximizes the consumer's utility. Plot this point on the graph, and label it as "Point X". Draw the indifference curve that goes through Point X, and label it "IC" (the indifference curve can be an approximation, it does not have to be exact). What is the slope of the indifference curve at Point X?

Question 27:

Adam derives utility from consuming apples and bananas. His marginal utility from consuming apples is

$$MU_A = 5A^4B^3,$$

while his marginal utility from consuming bananas is

$$MU_B = 3A^5B^2,$$

where A is the quantity of apples and B is the quantity of bananas. We know that the price of apples is $P_A = 6$, the price of bananas is $P_B = 2$ and Adam's income is $I=240$. Given this information, when Adam optimally chooses his consumption, he buys how many units of apples?

(A) A=65

(B) A=45

(C) A=25

(D) A=35

(E) A=55

Question 28:

Monica only consumes goods A and B. Suppose that her marginal utility from consuming good A is equal to $1/Q_A$, and her marginal utility from consuming good B is $1/Q_B$. If the price of A is \$0.50, the price of B is \$4.00, and the Monica's income is \$120.00, how much of good A will she purchase?

(A) 0

(B) 12

(C) 24

(D) 48

(E) 120